



CHAPTER

4

The Network Layer

We learned in the previous chapter that the transport layer provides various forms of process-to-process communication by relying on the network layer's host-to-host communication service. We also learned that the transport layer does so without any knowledge about how the network layer actually implements this service. So perhaps you're now wondering, what's under the hood of the host-to-host communication service, what makes it tick?

In this chapter, we'll learn exactly how the network layer implements the host-to-host communication service. We'll see that unlike the transport and application layers, there is a piece of the network layer in each and every host and router in the network. Because of this, network-layer protocols are among the most challenging (and therefore among the most interesting!) in the protocol stack.

The network layer is also one of the most complex layers in the protocol stack, and so we'll have a lot of ground to cover here. We'll begin our study with an overview of the network layer and the services it can provide. We'll then examine two broad approaches towards structuring network-layer packet delivery—the data-gram and the virtual-circuit model—and see the fundamental role that addressing plays in delivering a packet to its destination host.

In this chapter, we'll make an important distinction between the **forwarding** and **routing** functions of the network layer. Forwarding involves the transfer of a packet from an incoming link to an outgoing link within a *single* router. Routing

involves *all* of a network's routers, whose collective interactions via routing protocols determine the paths that packets take on their trips from source to destination node. This will be an important distinction to keep in mind as you progress through this chapter.

In order to deepen our understanding of packet forwarding, we'll look "inside" a router—at its hardware architecture and organization. We'll then look at packet forwarding in the Internet, along with the celebrated Internet Protocol (IP). We'll investigate network-layer addressing and the IPv4 datagram format. We'll then explore network address translation (NAT), datagram fragmentation, the Internet Control Message Protocol (ICMP), and IPv6.

We'll then turn our attention to the network layer's routing function. We'll see that the job of a routing algorithm is to determine good paths (equivalently, routes) from senders to receivers. We'll first study the theory of routing algorithms, concentrating on the two most prevalent classes of algorithms: link-state and distance-vector algorithms. Since the complexity of routing algorithms grows considerably as the number of network routers increases, hierarchical routing approaches will also be of interest. We'll then see how theory is put into practice when we cover the Internet's intra-autonomous system routing protocols (RIP, OSPF, and IS-IS) and its inter-autonomous system routing protocol, BGP. We'll close this chapter with a discussion of broadcast and multicast routing.

In summary, this chapter has three major parts. The first part, Sections 4.1 and 4.2, covers network-layer functions and services. The second part, Sections 4.3 and 4.4, covers forwarding. Finally, the third part, Sections 4.5 through 4.7, covers routing.

4.1 Introduction

Figure 4.1 shows a simple network with two hosts, H1 and H2, and several routers on the path between H1 and H2. Suppose that H1 is sending information to H2, and consider the role of the network layer in these hosts and in the intervening routers. The network layer in H1 takes segments from the transport layer in H1, encapsulates each segment into a datagram (that is, a network-layer packet), and then sends the datagrams to its nearby router, R1. At the receiving host, H2, the network layer receives the datagrams from its nearby router R2, extracts the transport-layer segments, and delivers the segments up to the transport layer at H2. The primary role of the routers is to forward datagrams from input links to output links. Note that the routers in Figure 4.1 are shown with a truncated protocol stack, that is, with no upper layers above the network layer, because (except for control purposes) routers do not run application- and transport-layer protocols such as those we examined in Chapters 2 and 3.